Milestone 1: Formula 1 Podium Prediction Companion

Analytical report - leakage-safe prediction after qualifying and before race start

The immutable 1950-2024 snapshot contains 26,759 raw result rows. A deterministic shared-drive rule resolves 85 duplicate driver-race groups, leaving 26,668 unique predictions.

Primary metric: PR-AUC, because podiums are a minority class. ROC-AUC is secondary; F1, precision, and recall use validation-selected thresholds.

Audit and cleaning

Literal \N values are parsed as missing. Dates and qualifying durations receive semantic types. All audited primary keys are unique and non-null, and all audited foreign keys reconcile. Raw CSV files are never mutated. For shared drives we prefer a row with grid > 0, then best final classification, then lowest resultId.

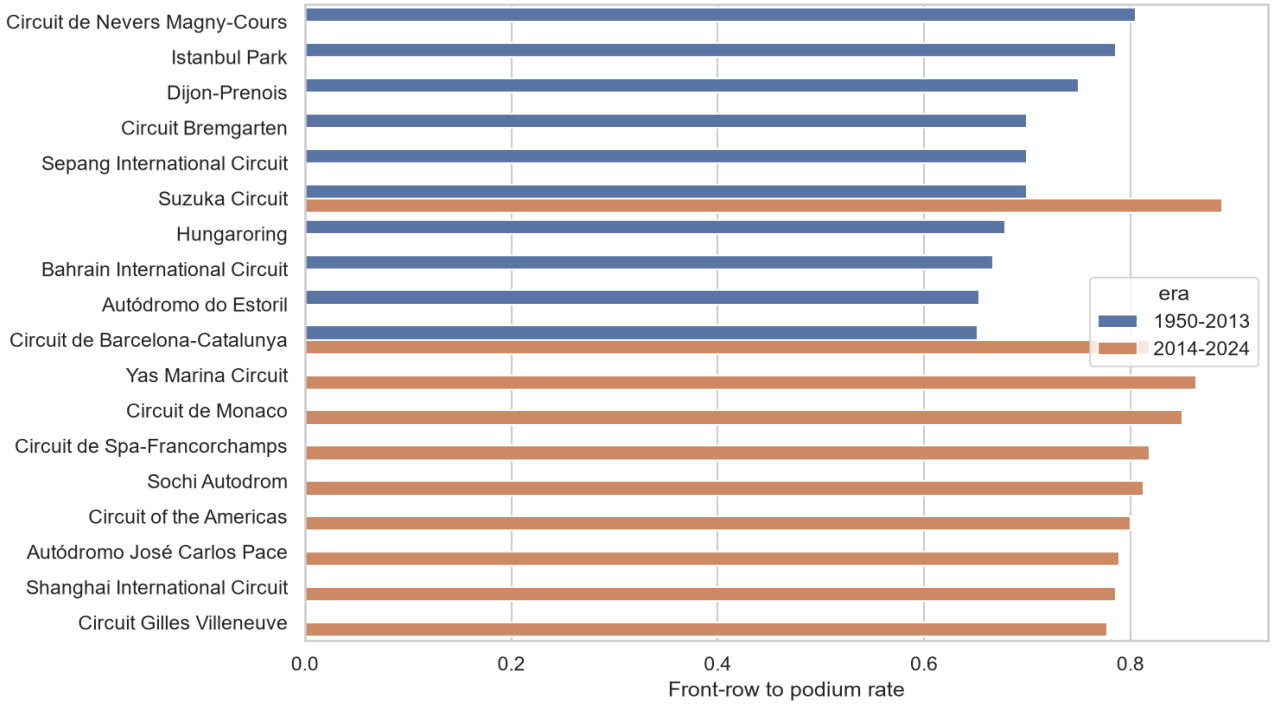


Qualifying coverage drifts strongly across history. The model therefore keeps an explicit missingness indicator, uses fitted imputation, and treats grid as the most consistent starting-order feature.

Data-engineering answers

1. Front-row conversion by circuit

A front-row start is defined by actual grid positions 1-2, not qualifying position, because penalties can change the grid. Results are joined through races to circuits and reported separately for pre-2014 and 2014-2024, with at least ten starts per circuit-era.

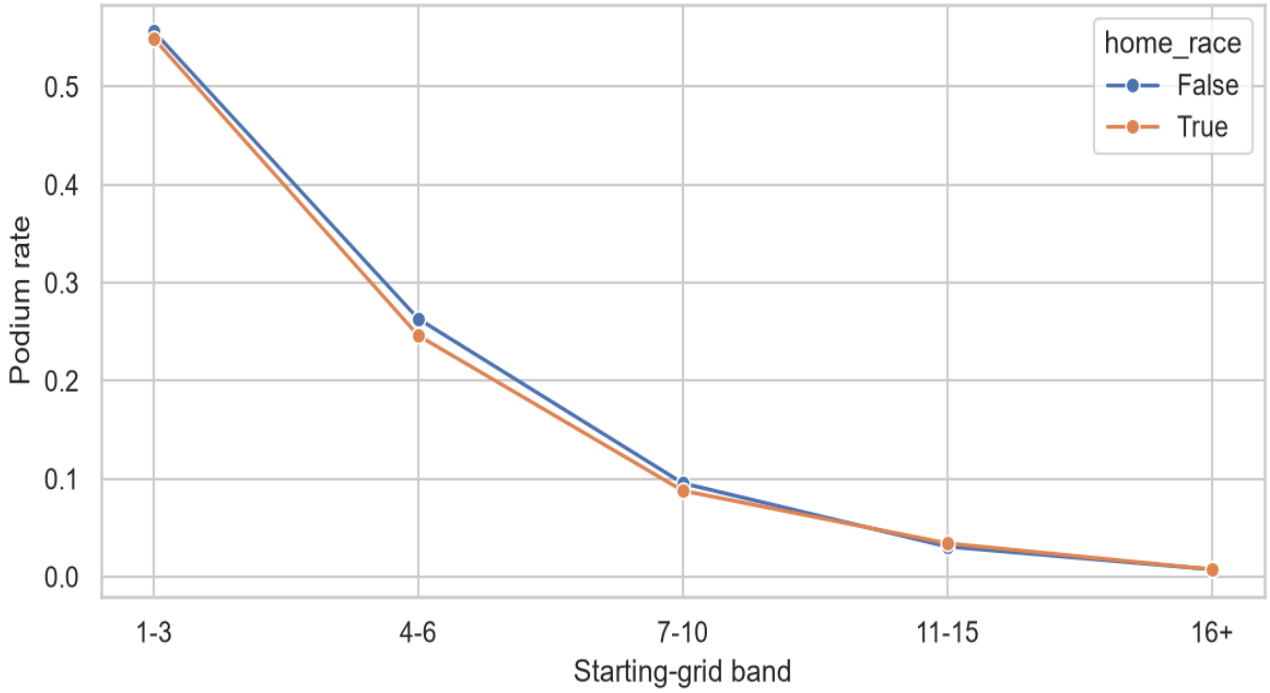


era	circuit_name	mean	count
1950-2013	Circuit de Nevers Magny-Cours	0.806	36.000
1950-2013	Istanbul Park	0.786	14.000
1950-2013	Dijon-Prenois	0.750	12.000
1950-2013	Circuit Bremgarten	0.700	10.000
1950-2013	Sepang International Circuit	0.700	30.000
2014-2024	Suzuka Circuit	0.889	18.000
2014-2024	Yas Marina Circuit	0.864	22.000
2014-2024	Circuit de Monaco	0.850	20.000
2014-2024	Circuit de Barcelona-Catalunya	0.818	22.000
2014-2024	Circuit de Spa-Francorchamps	0.818	22.000

Answer: the ranking changes across eras. Circuit de Nevers Magny-Cours leads 1950-2013 at 80.6%, while Suzuka Circuit leads 2014-2024 at 88.9% under the stated minimum-sample rule.

2. Home-race podium rate controlling for grid

Driver nationality is mapped to circuit country, then home and away podium rates are compared within the same starting-grid bands. This stratification controls the largest observed pre-race advantage without claiming a causal home effect.



grid_band	home_race	mean	count
1-3	False	0.557	3060.000
1-3	True	0.549	264.000
4-6	False	0.263	3016.000
4-6	True	0.246	260.000
7-10	False	0.095	3990.000
7-10	True	0.088	387.000
11-15	False	0.031	4933.000
11-15	True	0.034	469.000
16+	False	0.008	6904.000
16+	True	0.007	936.000

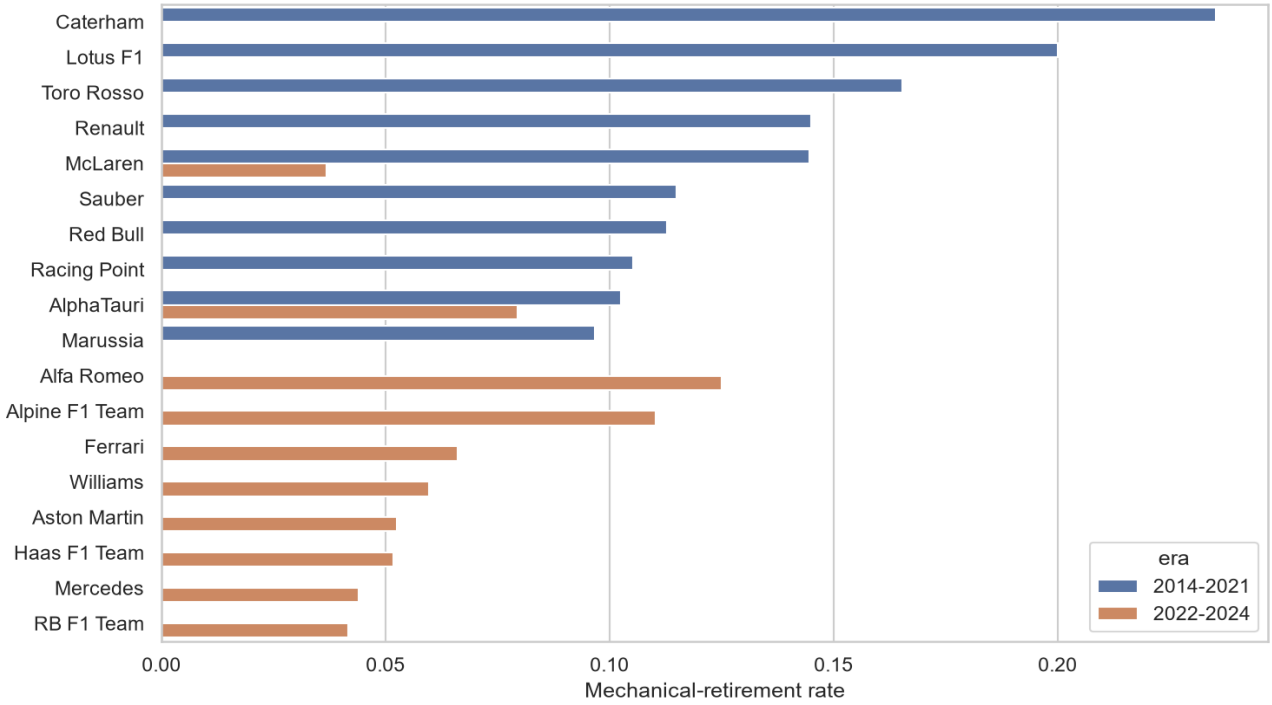
Answer: after comparing like starting-grid bands, home drivers do not show a consistent podium advantage. Their rates are slightly lower in four bands and only marginally higher in grid positions 11-15; this descriptive result does not establish causality.

3. Mechanical retirements

Results are joined to status and constructor. Only documented component failures count; crashes, driver errors, disqualifications, and regulation decisions are excluded. Rates use race starts as the denominator and require at least 20 starts per constructor-era.

Included mechanical statuses: Alternator, Battery, Brake duct, Brakes, Clutch, Cooling system, Differential, Driveshaft, Drivetrain, ERS, Electrical, Electronics, Engine, Exhaust, Front wing, Fuel leak, Fuel pressure, Fuel pump, Fuel system, Gearbox, Hydraulics, Mechanical, Oil leak, Oil pressure, Overheating, Power Unit, Power loss, Puncture, Radiator, Rear wing, Spark plugs, Steering, Suspension, Technical, Throttle, Transmission, Turbo, Tyre, Undertray, Vibrations, Water leak, Water pressure, Water pump, Wheel, Wheel bearing, Wheel nut.

Non-start statuses removed from the denominator: Did not prequalify, Did not qualify, Did not start, Excluded, Withdrew.



era	constructor_name	mean	sum	count
2014-2021	Caterham	0.235	8.000	34.000
2014-2021	Lotus F1	0.200	15.000	75.000
2014-2021	Toro Rosso	0.165	40.000	242.000
2014-2021	Renault	0.145	29.000	200.000
2014-2021	McLaren	0.145	46.000	318.000
2014-2021	Sauber	0.115	23.000	200.000
2014-2021	Red Bull	0.113	36.000	319.000
2022-2024	Alfa Romeo	0.125	11.000	88.000
2022-2024	Alpine F1 Team	0.110	15.000	136.000
2022-2024	AlphaTauri	0.080	7.000	88.000
2022-2024	Ferrari	0.066	9.000	136.000
2022-2024	Williams	0.060	8.000	134.000
2022-2024	Aston Martin	0.053	7.000	133.000
2022-2024	Haas F1 Team	0.052	7.000	135.000

Answer: Caterham has the highest mechanical-retirement rate in 2014-2021 (23.5%); Alfa Romeo leads 2022-2024 (12.5%). These rates use all starts, not only retirements, as the denominator.

Features and leakage control

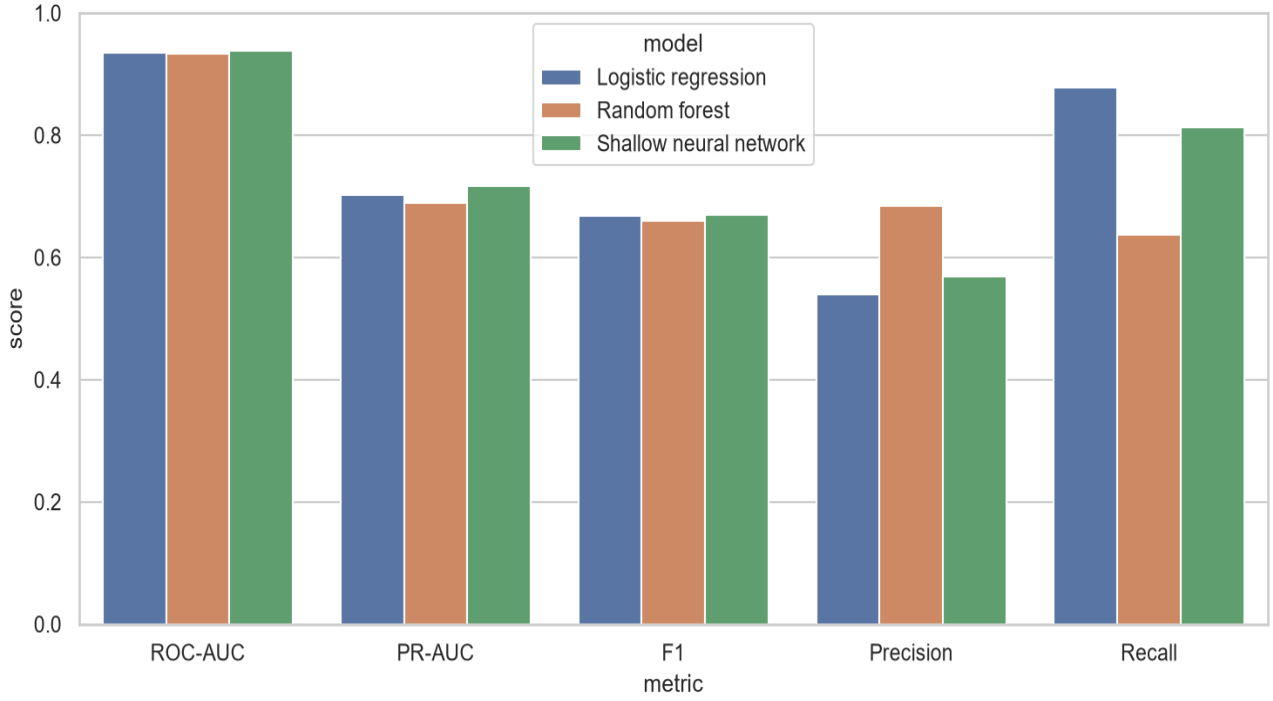
Inputs cover grid, qualifying, historical form, prior standings, and race context. Form is shifted before rolling, while standings after one race are attached only to the next race in that season. Current-race position, points, laps, status, pit stops, and fastest lap are excluded from the feature matrix. The chronological split is training through 2019, validation in 2020-2021, and unseen testing in 2022-2024.

Preprocessing experiments

preprocessing	validation_ROC-AUC	validation_PR-AUC
median + scaling	0.926	0.728
median, no scaling	0.924	0.724
zero + scaling	0.927	0.725

Model comparison

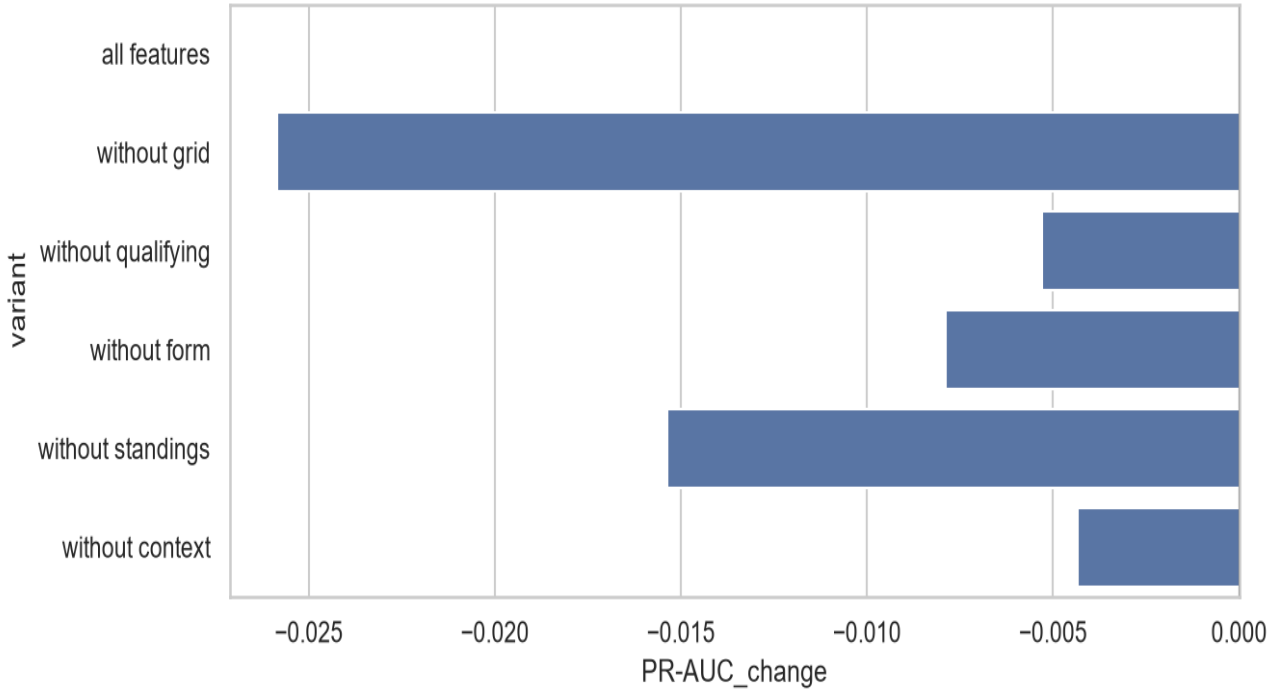
Logistic regression is the baseline; random forest tests nonlinear interactions; and the shallow MLP has two hidden layers (32 and 12 units). All models use the same rows and feature set. Hyperparameters and thresholds are chosen without consulting the test labels.



model	split	threshold	ROC-AUC	PR-AUC	F1	Precision	Recall
Logistic regression	train	0.655	0.906	0.561	0.580	0.468	0.763
Logistic regression	validation	0.655	0.926	0.728	0.703	0.647	0.769
Logistic regression	test	0.655	0.935	0.703	0.668	0.539	0.877
Random forest	train	0.640	0.988	0.919	0.829	0.827	0.831
Random forest	validation	0.640	0.924	0.702	0.709	0.757	0.667
Random forest	test	0.640	0.933	0.689	0.660	0.684	0.637
Shallow neural network	train	0.240	0.912	0.585	0.578	0.462	0.771
Shallow neural network	validation	0.240	0.928	0.738	0.713	0.677	0.752

model	split	threshold	ROC-AUC	PR-AUC	F1	Precision	Recall
Shallow neural network	test	0.240	0.938	0.716	0.669	0.568	0.814

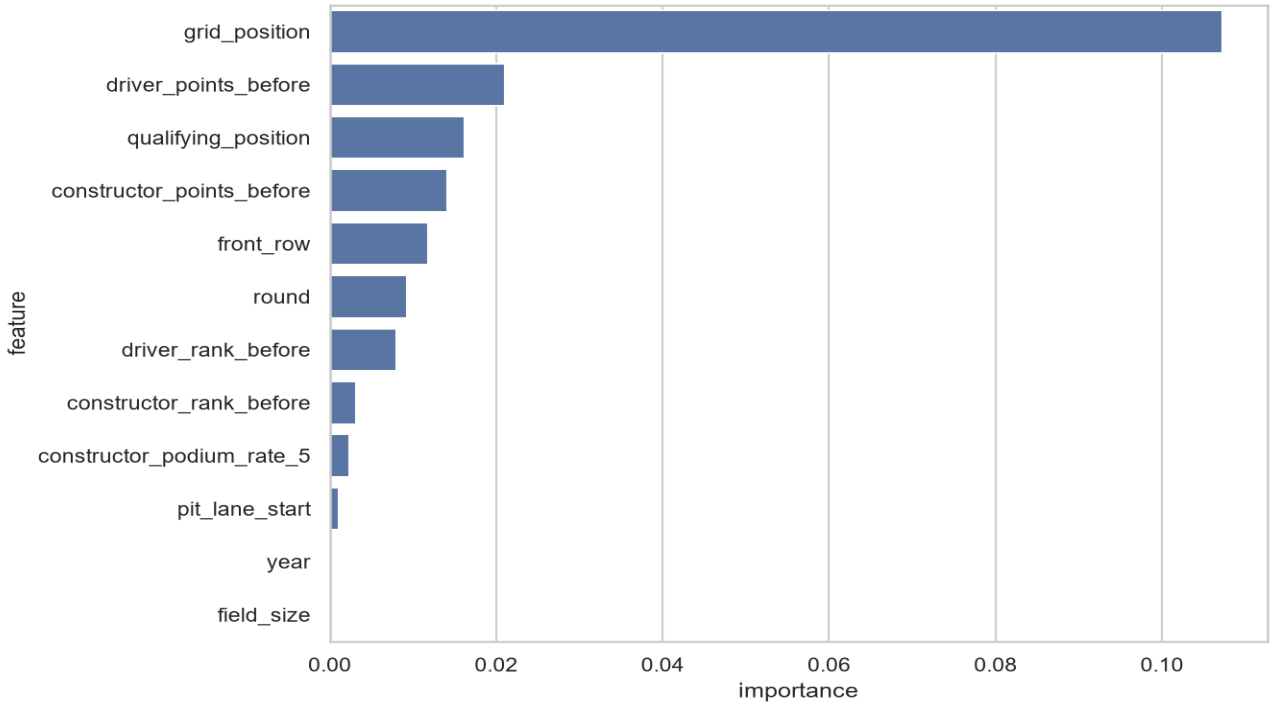
Feature-group ablation

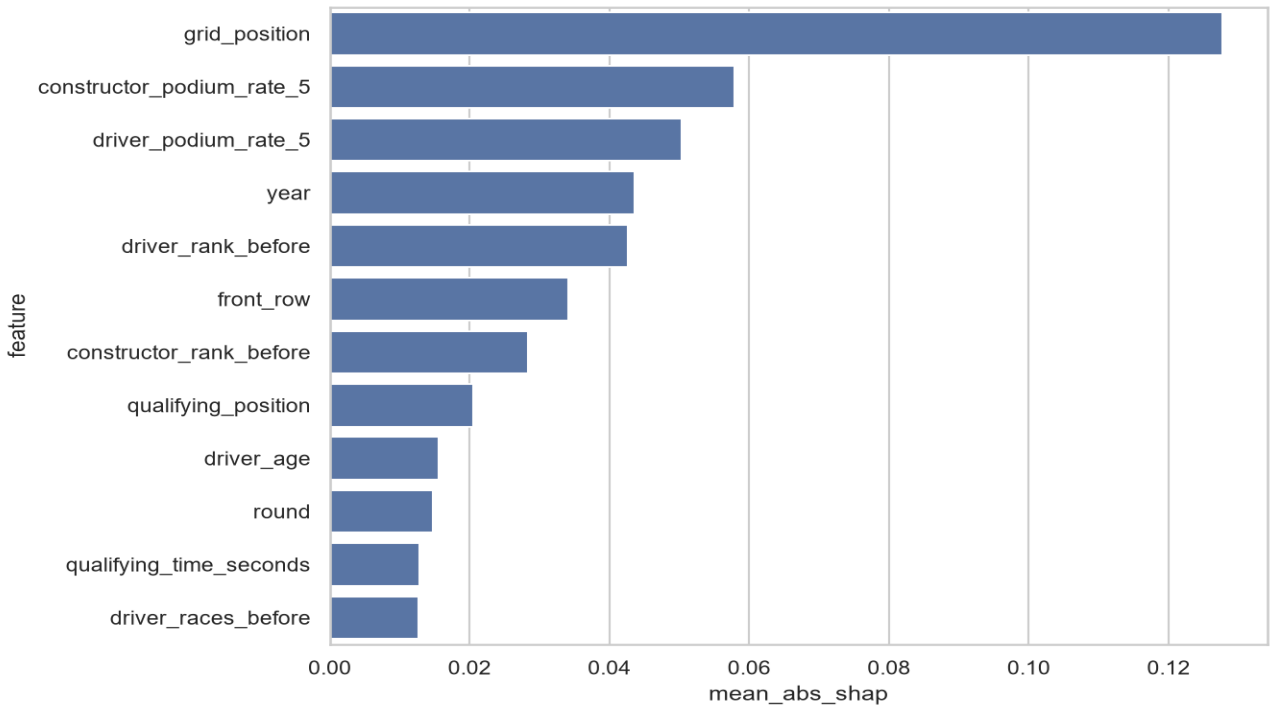


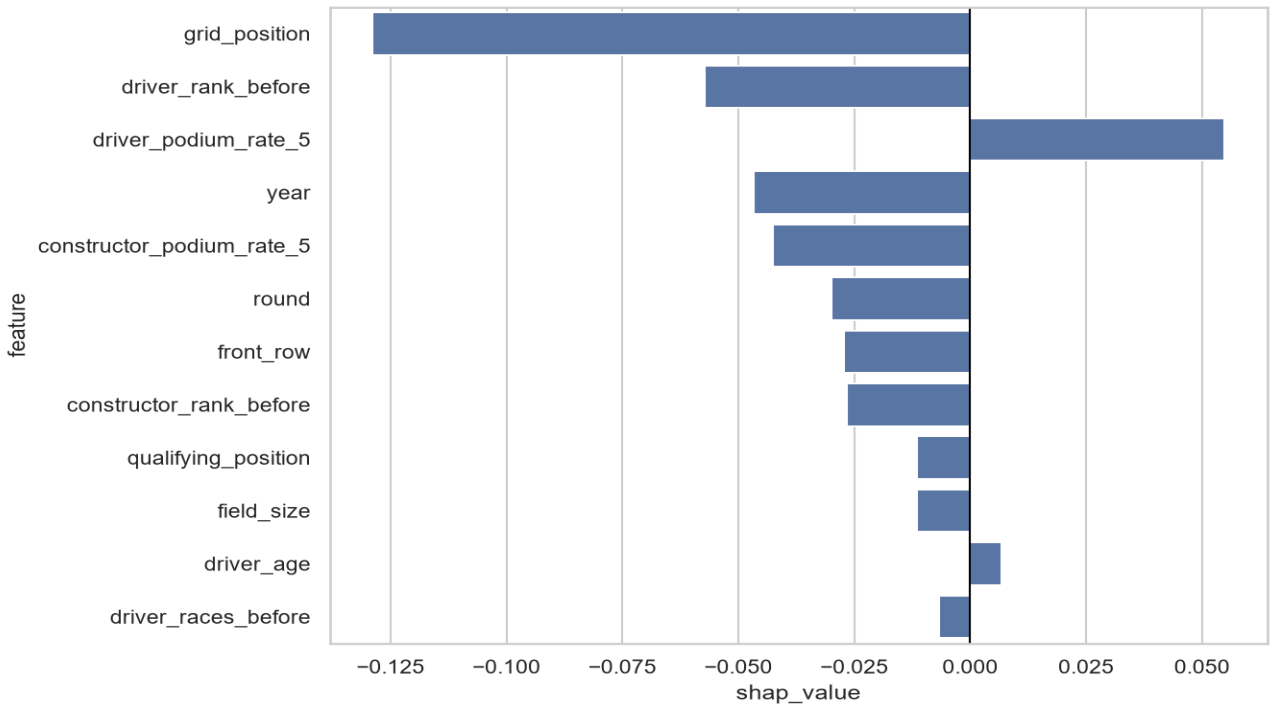
variant	features	validation_PR-AUC	PR-AUC_change
all features	17.000	0.728	0.000
without grid	13.000	0.702	-0.026
without qualifying	14.000	0.722	-0.005
without form	14.000	0.720	-0.008
without standings	13.000	0.712	-0.015
without context	14.000	0.723	-0.004

Explainability

Permutation importance and global SHAP describe model reliance across validation records; local SHAP explains one prediction. These are associations, not causal effects: importance does not prove that changing a feature causes a podium.







Inference and limitations

The reusable inference function was exercised with complete and partially missing records. The selected validation model was Shallow neural network; missing fields are handled by its training-fitted imputer.

Limitations include historical qualifying-coverage drift, constructor identity resets after renames, incomplete nationality mapping, conservative classification of ambiguous failure statuses, and era sensitivity. Predictions are probabilistic associations, not guarantees.